RADBox and RADBox_lite Setup Instructions

1. Download the Arduino IDE from the following link: https://www.arduino.cc/en/software. Make sure to carefully select the correct operating system before downloading, to ensure proper installation and compatibility with your system.

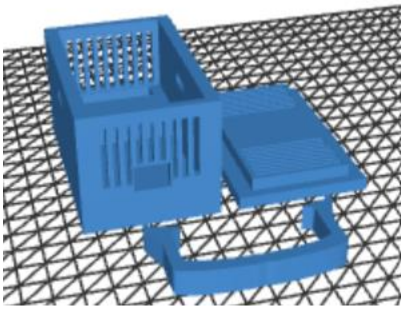
2. Download the code that will be uploaded to the Arduino. To do so, access the repository at https://github.com/Sofializarazo/RADbox and navigate to the source_code folder. For the RADBox_lite version, download the file RADBox_lite.ino. For the RADBox version, download the file RADBox.ino.
3. Open the Arduino application. It is important that you grant all requested permissions during setup and accept any extension installations that are prompted. These extensions are required to ensure that the code can be properly compiled and uploaded to the system.
4. Go to the upper-left corner of the Arduino IDE and click File, then select Open. Choose your .ino file. The program will prompt you to save the file in a folder with the same name. Click OK to accept and proceed.
5. Continue the instructions in “For RADBox_lite” or “For RADBox” depending on the case.

For Radbox_lite:

1. For the RADBox_lite system, the microcontroller being used is the Arduino Nano Every. This board is not included by default in the Arduino IDE, so it must be installed manually. To do it, go to the top menu in the Arduino IDE and click Tools > Board > Boards Manager. In the search bar, type “Arduino megaAVR Boards” and click Install. This package is required to enable support for the Arduino Nano Every and ensure the code can be properly compiled and uploaded
2. Now you must select the correct board configuration in the software: Go to Tools > Board >Arduino megaAVR Boards > Arduino Nano Every.
3. Then go to Tools > Registers Emulation and select “None (ATmega4809)”.

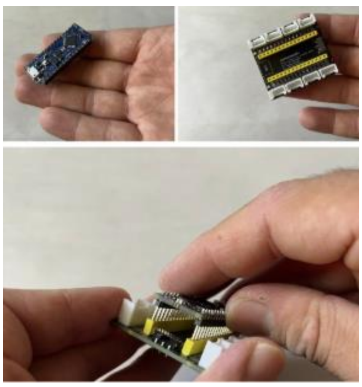
4. To ensure the code compiles without errors, it is necessary to install the SparkFun library. So go to Sketch >Include Library > Manage Libraries.... In the search bar, type "SparkFun SCD4x Arduino Library" and install it.
5. Connect the Arduino to your computer using a Micro-USB cable.
6. Then go to Tools > Port and select the port where the Arduino is connected.
7. Upload the code by clicking the right-arrow (Upload) button in the top toolbar. Wait until the bottom status bar displays the message "Done uploading".
8. Your Arduino is now ready to use.

9. Download the STL file "The_RADBox_Lite_FINAL.stl" from the enclosure_files folder in the RADBox GitHub repository. This file contains the enclosure model for RADBox_lite, which is designed to house electronic circuitry. The model has been fully prepared and optimized for fabrication and is ready to be printed using a commercial 3D printer.



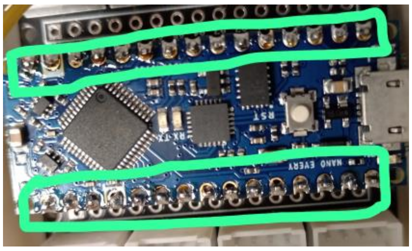
RADBox_lite design

10. To begin assembling the system, plug the Arduino Nano Every into the Grove Shield for Arduino Nano. Ensure that the Arduino Nano Every is inserted in the correct orientation by aligning it with the USB port diagram printed on the Grove Shield for Arduino Nano. Incorrect installation may prevent the system from functioning properly.



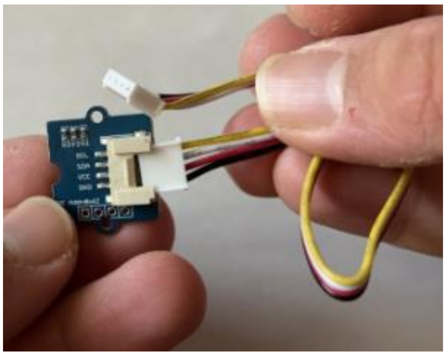
Arduino Nano Every (left) and *Grove Shield for Arduino Nano* (right) connecting (bottom).

11. This is not really necessary but, if it is possible solder the Arduino Nano directly to the Grove Shield for Arduino Nano board to prevent any movement from causing a disconnection in the RADBox_lite system and to ensure a more stable and reliable connection. During the soldering process, make sure that each solder joint remains isolated from the others, as accidental bridges between adjacent contacts may result in a short circuit.



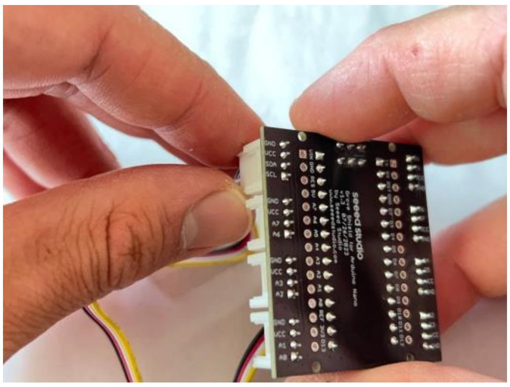
Arduino Nano soldered directly to the Grove Shield for Arduino Nano board

12. Connect one end of a universal 4 pin connector to the CO_2 sensor.



Grove CO2 sensor connected to a universal 4 pin cable.

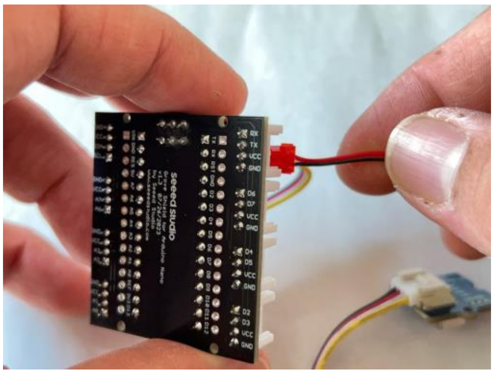
13. Plug the cable with the CO_2 sensor into the I2C port on the Grove Shield for Arduino Nano (labelled "GND VCC SDA SCL").



Grove CO_2 sensor plugged into the Grove Shield for Arduino Nano

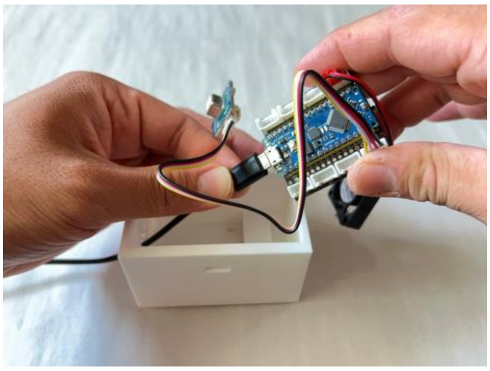
14. Plug the brushless fan into any other port on the Grove Shield for Arduino Nano. Ensure the black wire connects to a pin labelled

"GND" and the red wire connects to the pin on the same port labelled "VCC".



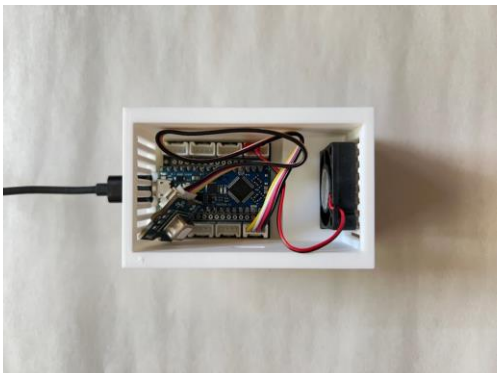
Grove CO_2 sensor plugged into the Grove Shield for Arduino Nano

15. Run the micro-usb connector through the cable port hole in the RADBox Lite and connect to the Arduino Nano Every.



Arduino Nano Every plugged into micro-usb.

16. Place the Arduino Nano Every and the brushless fan in the appropriate slots in the RADBox Lite. The fan will blow out of the side of the fan with the label and may be placed blowing in either direction.



Internals of assembled RADBox Lite.

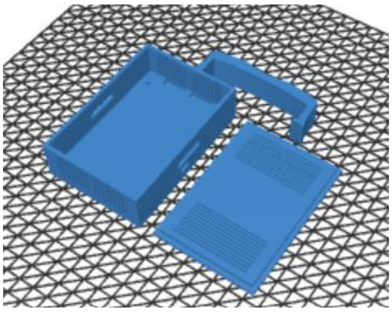
17. Place the lid onto the RADBox Lite and secure the clip. The RADBox Lite is now assembled.

Fully assembled RADBox Lite.

18. If you would like to see a video of the RADBox_lite assembly, please follow this link: RADBox Lite Assembly Instructions

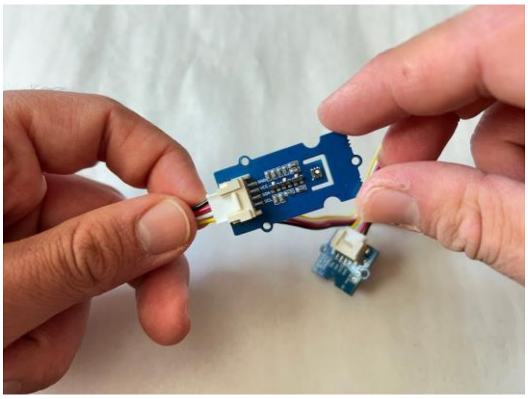
For Radbox:

1. To ensure the code compiles without errors, it is necessary to install some libraries So go to Sketch >Include Library > Manage Libraries.... In the search bar, type "SparkFun SCD4x Arduino Library" and install it, then type "SensirionI2CSgp41" and install it, then the same with "DFRobot_DHT20" and finally type "U8g2" and install the Oliver version.
2. Connect the Arduino to your computer using a Micro-USB cable.
3. Go to Tools> Board and select Arduino Uno.
4. Then go to Tools > Port and select the port where the Arduino is connected.
5. Upload the code by clicking the right-arrow (Upload) button in the top toolbar. Wait until the bottom status bar displays the message "Done uploading".
6. Your Arduino is now ready to use.
7. Download the STL file "The_RADBox_FINAL.stl" from the enclosure_files folder in the RADBox GitHub repository. This file contains the enclosure model for RADBox, which is designed to house electronic circuitry. The model has been fully prepared and optimized for fabrication and is ready to be printed using a commercial 3D printer.



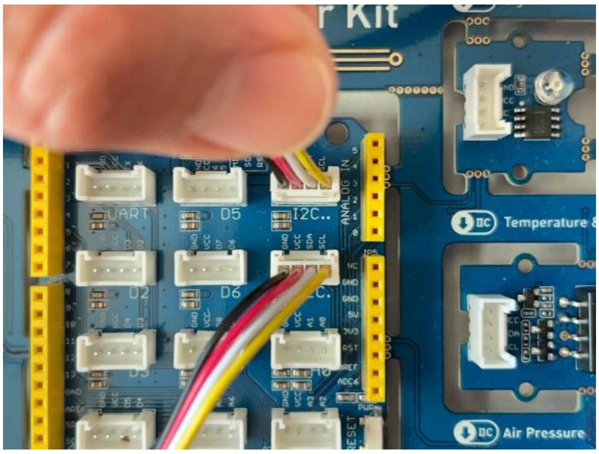
RADBox design

8. Connect one end of a universal 4 pin connector to the CO_2 and VOC/NO_x sensor.



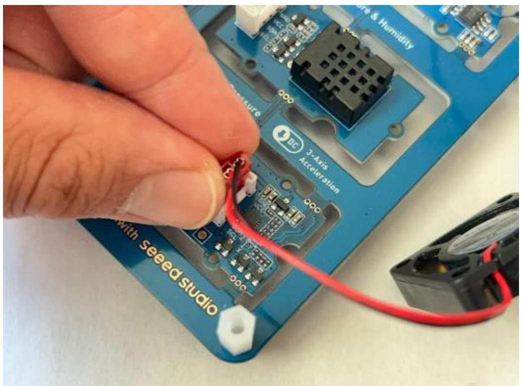
Grove CO_2 and VOC/NO_x sensor connected to a universal 4 pin cable.

9. Ensure the cables with the CO_2 and VOC/NO_x sensors are plugged into the I2C ports on the Grove Beginner Kit For Arduino (labelled "GND VCC SDA SCL").



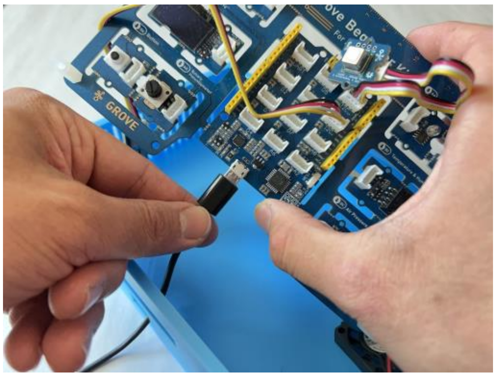
Grove CO2 and VOC/NOx sensor plugged into the Grove Beginner Kit For Arduino.

10. The brushless fans may be plugged into any other port on the Grove Beginner Kit For Arduino. Ensure the black wire connects to a pin labelled "GND" and the red wire connects to the pin on the same port labelled "VCC".



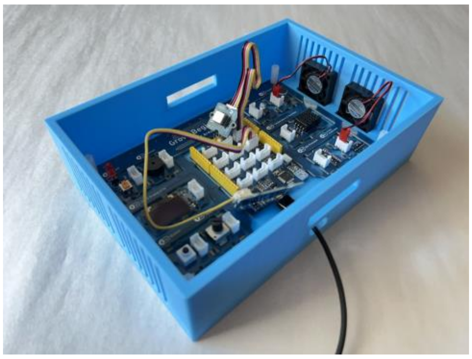
Brushless fan plugged into the Grove Beginner Kit For Arduino.

11. Run the micro-usb connector through the cable port hole in the RADBox and connect to the Grove Beginner Kit For Arduino.



Grove Beginner Kit For Arduino plugged into micro-usb.

12. Place the Grove Beginner Kit For Arduino and the brushless fans in the appropriate slots in the RADBox. The fans will blow out of the side of the fan with the label and may be placed blowing in either direction. However, ensure they are both orientated facing the same direction.



Internals of assembled RADBox.

13. Place the lid onto the RADBox and secure the clip. The RADBox is now assembled.

Fully assembled RADBox.

14. If you would like to see a video of the RADBox assembly, please follow this link:[RADBox Assembly Instructions]